

Calculus 2 - Homework 4 (Worked Solutions)

1.

$\Delta x = (4 - 0)/4 = 1$. Nodes: 0, 1, 2, 3, 4.

$f(x) = \sqrt{x}$: $f_0=0$, $f_1=1$, $f_2=1.4142$, $f_3=1.7321$, $f_4=2$.

Simpson: $S = (\Delta x/3)[f_0 + 4f_1 + 2f_2 + 4f_3 + f_4]$.

$S = (1/3)[0 + 4(1) + 2(1.4142) + 4(1.7321) + 2]$

$= (1/3)[0 + 4 + 2.8284 + 6.9282 + 2] = (1/3)(15.7566) = 5.2522$.

(Exact value $(2/3) \cdot 4^{3/2} = 16/3 = 5.3333$; sqrt has infinite slope at 0, so Simpson is a bit low.)

ANSWER: $S_4 = 5.2522$

2.

The integrand blows up at $x = 0$, so this is improper at the lower limit.

$\int_0^1 x^{-1/2} dx = \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} dx$.

$\int x^{-1/2} dx = 2x^{1/2}$, so $\int_t^1 x^{-1/2} dx = 2 - 2\sqrt{t}$.

As $t \rightarrow 0^+$, $\sqrt{t} \rightarrow 0$, giving the limit 2.

Converges (p-integral at 0 with $p = 1/2 < 1$).

ANSWER: converges to 2

3.

The 2^n suggests a ratio test, but a direct comparison is immediate.

For all $n \geq 1$, $2^n + n > 2^n$, so $0 < 1/(2^n + n) < 1/2^n = (1/2)^n$.

$\sum (1/2)^n$ is a convergent geometric series (ratio $1/2 < 1$).

By Direct Comparison, $\sum 1/(2^n + n)$ converges.

ANSWER: converges (direct comparison with geometric sum $(1/2)^n$)

4.

Use the double-angle identity $\sin x \cos x = (1/2) \sin 2x$, so

$\sin^2 x \cos^2 x = (1/4) \sin^2(2x)$.

Then power-reduce: $\sin^2(2x) = (1 - \cos 4x)/2$, giving

$\sin^2 x \cos^2 x = (1/8)(1 - \cos 4x)$.

$\int_0^{\pi/2} (1/8)(1 - \cos 4x) dx = (1/8)[x - \sin(4x)/4]_0^{\pi/2}$.

At $\pi/2$: $(\pi/2 - \sin(2\pi)/4) = \pi/2 - 0 = \pi/2$. At 0: 0.

Result = $(1/8)(\pi/2) = \pi/16$.

ANSWER: $\pi/16$

5.

For all $n \geq 1$, $n^2 + 1 > n^2$, so $0 < 1/(n^2 + 1) < 1/n^2$.

The series $\sum 1/n^2$ is a convergent p-series ($p = 2 > 1$).

By the Direct Comparison Test, the smaller nonnegative series

$\sum 1/(n^2 + 1)$ also converges.

ANSWER: converges (direct comparison with $\sum 1/n^2$)

6.

We use the substitution $u = x^2 + 1$.

Then $du = 2x dx$, which is exactly the $2x dx$ already present.

Rewrite the integral in u : $\int u^5 du$.

Apply the power rule: $\int u^5 du = u^6/6 + C$.

Back-substitute $u = x^2 + 1$: $(x^2 + 1)^6/6 + C$.

Check: $d/dx [(x^2+1)^6/6] = 6(x^2+1)^5(2x)/6 = 2x(x^2+1)^5$, the integrand.

ANSWER: $(x^2 + 1)^6 / 6 + C$

7.

Form $(a^2 + x^2)$ with $a = 2$. Substitute $x = 2 \tan \theta$,
 $dx = 2 \sec^2 \theta d\theta$, and $x^2 + 4 = 4 \sec^2 \theta$ so
 $(x^2 + 4)^{3/2} = (4 \sec^2 \theta)^{3/2} = 8 \sec^3 \theta$.
 $\int (2 \sec^2 \theta) / (8 \sec^3 \theta) d\theta = (1/4) \int \cos \theta d\theta$
 $= (1/4) \sin \theta + C$.
From $x = 2 \tan \theta$: $\sin \theta = x / \sqrt{x^2 + 4}$.
Result $= (1/4)(x / \sqrt{x^2 + 4}) + C = x / (4 \sqrt{x^2 + 4}) + C$.
ANSWER: $x / (4 \sqrt{x^2 + 4}) + C$

8.

$\int_0^\infty e^{-2x} dx = \lim_{t \rightarrow \infty} \int_0^t e^{-2x} dx$.
 $\int e^{-2x} dx = -(1/2) e^{-2x}$, so $\int_0^t e^{-2x} dx = -(1/2) e^{-2t} - (-(1/2))$
 $= 1/2 - (1/2) e^{-2t}$.
As $t \rightarrow \infty$, $e^{-2t} \rightarrow 0$, so the limit is $1/2$.
ANSWER: converges to $1/2$

9.

Check absolute convergence first: $\sum |(-1)^n / n^2| = \sum 1/n^2$ is a
convergent p-series ($p = 2 > 1$).
Since the series of absolute values converges, the original series is
absolutely convergent (and therefore convergent).
ANSWER: absolutely convergent

10.

Odd power of sine: split off one sine, convert the rest with
 $\sin^2 x = 1 - \cos^2 x$.
 $\int \sin^3 x \cos^2 x dx = \int (1 - \cos^2 x) \cos^2 x \sin x dx$.
Let $u = \cos x$, $du = -\sin x dx$.
 $= \int (1 - u^2) u^2 (-du) = \int (u^4 - u^2) du = u^5/5 - u^3/3 + C$.
Back-substitute $u = \cos x$: $\cos^5 x / 5 - \cos^3 x / 3 + C$.
ANSWER: $\cos^5(x)/5 - \cos^3(x)/3 + C$